Fiscal Year:	FY 2014	**Task Last Updated:**	FY 06/13/2014
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Project Title:	Kinetics of Electric Field-Driven Phase Transitions in Polarized Colloids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	07102-1982	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Task Description:	Motivation: Colloidal assembly provides the capability for scalable manufacturing of structured materials and devices for traditional and emerging industries. Although equilibrium structures and thermodynamics of colloids exposed to a field have been well studied, the use of electrically assisted assembly is limited by a lack of understanding of the role of time and spatially varying fields in directing and controlling non-equilibrium processes. Microgravity offers an opportunity to study field-driven phenomena on non-buoyancy-matched colloids by removing masking gravity effects, such as particle sedimentation, convection, and jamming. Objectives: Design and conduct microgravity tests in the International Space Station Advanced Colloids Experiment facility in long timescales to elucidate fundamental microscopic mechanisms of non-equilibrium phenomena underlying the field-assisted assembly of interacting colloidal particles and suggest novel routes for creating functional materials. The emphasis will be on the field-driven evolution of the mesoscale phase transitions, electro-hydrodynamic instabilities, and nucleation and growth of crystalline structures. The phenomena selected are logical candidates to take advantage of microgravity because terrestrial observations have failed to provide a comprehensive physical understanding of the underlying physics. Methodology: Dielectrophoretic experiments will be carried out on well-characterized colloids under well-defined conditions in terrestrial and microgravity environments. We will use electro-hydrodynamic methods and apparatuses for the control and manipulation of particles which were developed in our NASA-supported projects, including parabolic flight tests conducted in 2002 and planned in the summer of 2013. Molecular dynamics simulations, coarse-grained and continuum models will be employed to study field-induced colloidal structures. Deliverables: Detailed understanding of electrically driven non-equilibrium phenomena in colloids, apparatuses for the field-assisted colloid assembly, experimental database, and predictive models for the field-induced structure. The research will be integrated with educational and outreach initiatives in space sciences and related technologies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research has the strong potential to reveal the fundamental microscopic mechanisms of electrically driven phenomena in a suspension of interacting particles and develop a physical explanation for puzzling observations under terrestrial conditions that are not predicted by current theories. The main focus is on the appearance of a cellular structure in a colloid exposed to a spatially uniform field and understanding of the nucleation and growth of crystalline structures in a high-gradient and mechanisms of electro-hydrodynamic instabilities of concentration fronts. The obtained results will facilitate the development of new technologies for the control and manipulation of colloidal processes. Space applications: Electrical technologies can be employed in space to mitigate the effects of reduced gravity by providing an electrical body force. Results of ACE-T-E experiments will be useful for the development of new methods for electro-hydrodynamic control, manipulation, and separation of multiphase fluid flows in various space applications, for example, thermal management systems. Earth applications: The research should bring the understanding of electrically driven colloidal processes to the level useful in the design of apparatuses and processes for a broad range of applications in materials engineering, life science, electronics, and chemical industry. This will stimulate further "spin-offs" of electrical technologies.
Task Progress:	Major Accomplishments The study was conducted within the scope of the originally proposed research plan. The project team worked in closed collaboration with NASA projects scientists and teams conducting ISS ACE investigations. The main focus was on the development of science requirements for experiments aboard the ISS. Ground-based experiments were conducted to select cell geometry, electrode configuration, voltage/frequency regimes and suspensions for the ISS tests. Hardware: The ACE-T-E experiment will utilize the hardware in the Fluids Integrated Rack (FIR) aboard ISS. Existing capabilities include epi-illumination, simple LED back-lighting, and oil immersion microscopy using up to 100x magnification, white light illumination, fluorescent tagging of samples, and monochrome and video imaging. Existing FIR capabilities provide data acquisition and control, sensor interfaces, laser and white light sources, advanced imaging capabilities, power, cooling, and other resources. An experimental unit will be assembled in the ACE-T-E module that will provide control of the sample temperature and AC high-voltage to energize electrodes of specialty cells. To quantify electrically driven colloidal processes, the imaging system will record the field-driven evolution of the particle distribution in the specialty cell. The specialty cells will be developed to study the electric field-induced phase transitions, dynamics of concentration fronts, electro-hydrodynamic instabilities and formation of crystalline structures in colloids. An experimental unit, each loaded with a particular system, will comprise a specialty cell (1 - 2 mL) equipped with transparent electrodes, a container (100 mL desirable) and a pump connected in a single closed loop. Materials: Experiments will be conducted on relatively concentrated suspensions (0.5-10 v/v %) of non-buoyancy-matched polarized particles (1-10 micrometers). We expect to conduct experiments on 4 different types of particles (several concentrations for each type of particles) to cover a wide range of the mismatch of dielectric constants and conductivities between particles and a suspending liquid. The particle type, the suspending liquid and the number and volume of samples of different concentrations within each type may change, as specialty cell designs, high-voltage unit and payload availability clarified. Operational requirements: • The unit will be fully filled with a colloid in the Earth facility. As the particles will sediment in Earth’s gravity and the hyper-gravity during the launch, pumping the colloid through the unit for a certain period of time will be conducted aboard the ISS to homogenize it prior to observation. If needed, a magnetic stirrer will be used to homogenize a colloid in the container. • A sample in the specialty cell will be heated to a desired temperature and then exposed to an electrical field by

	applying a desired voltage/frequency for a certain period of time. The imaging system will record the field-driven evolution of the colloidal structure. The duration of a particular experiment will be determined in real-time as data sets are transmitted to Earth. It cannot be predicted based on observations of the field induced behavior of non-buoyancy-matched suspensions under Earth's gravity. • After the exposure of the sample to the field, the pump will be turned on to flow the sample from the specialty cell to the container and simultaneously the fresh sample from the container into the specialty cell. • To quantify the field effects on colloidal processes, experiments on each colloid will be carried out at different voltages, frequencies and temperatures. • The unit needs to be sealed in order to repeat experiments later for different voltage/frequency and temperature regimes. • Each colloid will be loaded in its particular unit. • For each colloid, 3 - 6 magnitudes of high voltage will be tested at each selected temperature; several tests at each voltage for evaluating reproducibility; at least, two different temperatures are desirable. • At each voltage, experiments will be conducted at several frequencies; duration will be determined in real-time as data sets are transmitted to Earth. • A sample in the specialty cell will be replaced with a fresh sample from the container after exposure to an electric field. • Analyze data after each exposure of a colloid to a fixed voltage/frequency and temperature regime as data sets are transmitted to Earth and, depending on the results, re-write scripts and adjust parameters of the following tests.
Bibliography Type:	Description: (Last Updated: 02/06/2025)
Abstracts for Journals and Proceedings	Khusid B, Löwen H, Nelson ES, Sheehan CC, Telo da Gama MM, Yethiraj A. "Electric field-driven phase transitions in polarized colloids." Presented at the 29th Annual Meeting of the American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, Florida, November 3–8, 2013. 29th Annual Meeting of the American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. Program and abstracts, p. 117. , Nov-2013